# FIN ST432–ST446

## Sector Exclusion, a Certified Stationary-Orbit Census, and Local Infrared Degree

Release 10.86

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**Abstract**

Programs ST432–ST446 continue the local analytical and computational FIN program after Release 10.85. The main global-variational advance is a rigorous two-sector exclusion at every supplied gain $g \le 2.88$. No negative competitor to the uniform state exists when $\max_i p_i \le 0.14824732498$ or when $\max_i p_i \ge 0.8$; any missed competitor must lie in the explicit intermediate band. This does not close the global transition. The already certified reflection-even branch crosses the uniform energy transversely in the narrow ST342 interval, with $dV/dg \in [-0.671281367, -0.671275547]$, but exclusion of a third competitor remains open.

The singular infrared continuation is strengthened substantially. One common parametric Krawczyk box now proves a unique compactified root for every $b = 1/y_3 \in [0, 0.017]$, an 8.5-fold extension. On $b \in [0, 0.015]$ a uniform contraction certificate fixes the Jacobian orientation and proves local Brouwer degree $-1$. Separately, symmetry-averaged interval boxes determine the exact stabilizers of nine previously isolated stationary representatives. Their $D_{12}$ orbits contain 85 distinct certified stationary points. The partial Morse polynomial is $13 + 42t + 30t^2$ and is already Euler-balanced, $13 - 42 + 30 = 1$, although completeness and heteroclinic connections are not proved.

The batch also excludes all degree-eight invariant syzygies of support at most two over two primes, closes the characteristic-zero degree-five primitive quotient at dimension 52, improves synthetic E- and A-optimal design metrics with one exchange, sharpens a numerical transfer-gap sequence, and enlarges a conservative local input-to-state stability ball. No result derives the gain, selects one orbit member, supplies dimensional units, completes a legacy-to-strict physical-role transfer, or adds external empirical evidence.

# Contents

## 1 Scope, source packet, and nonclaims

The audited repository baseline is Git commit `bb540a767e210a5a3bf816e4868e7bcefbac3872`, supplemented by the uncommitted local ST417–ST431 packet and the executable artifacts listed in the reproducibility section. The strict finite operator is

$$A = sI - W, \qquad W_{xx} = 0, \qquad W_{xy} = \frac{\cos(0.18575d(x,y) + 0.16250)}{1 + d(x,y)^{9/5}} \quad (x \neq y), \tag{1}$$

on $\mathbb{Z}_{12}$, with off-diagonal row sum $s = 1.660307278766099\ldots$. The supplied variational family is

$$V_g(p) = D(p\|u) - \frac{g}{2}q^T Aq, \qquad q = p - u, \quad u_i = \frac{1}{12}, \quad D(p\|u) = \sum_i p_i \log(12p_i). \tag{2}$$

All quantities below are dimensionless.

The legacy ontological kernel remains an intermediate bridge kernel. It is not silently identified with (1); no legacy electroweak, electromagnetic, gravity-hierarchy, Planck-scale, or other physical role is transferred. The project ontology is retained only as context: the nadsoliton itself is primordial fractal information in a solitonic state—there is no separate informational layer below it—with the preferred internal order nadsoliton → light → matter → emergent observer.

A $D_{12}$ orbit is not a selector. QW-2191 remains open. A theorem conditional on $g$ does not source $g$. A dimensionless optimization branch is not a physical infrared phenomenon, and local computation is not external validation.

## 2 Executive result ledger

| Program | Status | Result |
|---|---|---|
| ST432 | **[PROVEN]** limited | Over each of two primes, all 2028 degree-eight candidates are nonzero and pairwise nonproportional; support-one/two syzygies are excluded. The forced 136-dimensional relation problem remains unsolved. |
| ST433 | **[PROVEN]** partial | For $g \leq 2.88$, negative competitors are excluded in the uniform and concentrated outer sectors; the band $0.1482473 < \max p_i < 0.8$ remains open. |
| ST434 | **[PROVEN]** local | The certified reflection-even equal-value crossing is transverse and locally unique in gain; global third-competitor exclusion is absent. |
| ST435 | **[PROVEN]** nonexhaustive | Exact stabilizers and orbit sizes certify 85 distinct stationary points generated by the nine isolated representatives. |
| ST436 | **[PROVEN]** census / **[PARTIAL]** topology | The partial Morse polynomial is $13 + 42t + 30t^2$ and has Euler sum one. Completeness and Conley connections remain open. |
| ST437 | **[PROVEN]** | A common Krawczyk box proves unique singular attachment for every $b \in [0, 0.017]$. |
| ST438 | **[PROVEN]** local | On $b \in [0, 0.015]$, $\|I - CJ\|_\infty \leq 0.938101 < 1$ and the branch component has Brouwer degree $-1$. |
| ST439 | **[PROVEN]** bounded | The characteristic-zero degree-five primitive quotient has dimension 52; degree six is bounded between 39 and 55, not closed. |
| ST440 | **[STRONG NUMERICAL EVIDENCE]** | One weakest-direction row exchange improves the synthetic E metric by 4.29% and the A metric by 4.16%. |
| ST441 | **[PROVEN]** cone bound / **[STRONG NUMERICAL EVIDENCE]** spectrum | Five Ulam grids stabilize near a modulus ratio 0.595740; no infinite-dimensional two-sided gap bracket is proved. |

| Program | Status | Result |
|---|---|---|
| ST442 | **[PROVEN]** conditional | A paid local ISS ball has radius $1.93 \times 10^{-6}$ and forcing threshold $4.2163 \times 10^{-5}$. |
| ST443 | **[BLOCKED]** | Mixed rank-seven edge signs still defeat the positive-weight envelope; no certified sign-aware SDP closure is obtained. |
| ST444 | **[BLOCKED]** | No internal strict source of $g$ is exported. |
| ST445 | **[BLOCKED]** | No nonpremise selector is exported; QW-2191 remains open. |
| ST446 | **[BLOCKED]** | No independent referee, laboratory record, custody chain, or held-out empirical datum is added. |

# 3 ST433–ST434: what is now known about the gain transition

For $p \neq u$ with $q^T Aq > 0$, equality $V_g(p) = V_g(u) = 0$ occurs at

$$g(p) = \frac{2D(p\|u)}{q^T Aq}. \tag{3}$$

Consequently the first possible global change satisfies the exact variational identity

$$g_c = \inf_{p \neq u} \frac{2D(p\|u)}{q^T Aq}. \tag{4}$$

This reformulates the unresolved problem as a global ratio minimization; it does not assume the known local branch is the minimizer of (4).

## 3.1 Uniform sector

Let $\lambda_{\max}(A) = 2.342182041146301\ldots$. If $\max_i p_i \leq m$, then on the tangent space

$$\nabla^2 V_g(p) = \mathrm{diag}(1/p_i) - gA \succeq (1/m - g\lambda_{\max})I.$$

Thus the convex sector containing $u$ has no negative competitor whenever $m \leq 1/(g\lambda_{\max})$. At $g = 2.88$ this gives

$$\max_i p_i \leq 0.14824732498259885. \tag{5}$$

## 3.2 Concentrated sector

Write the largest coordinate as $1 - t$, normalize the other eleven to a vector $y$, and put $\rho = \sum_j y_j^2$. Entropy extremization at fixed $\rho$, a feasible SOCP-dual water-fill lower bound for the center-to-remainder energy, and the smallest strict edge weight give a two-variable lower envelope. A $1000 \times 1000$ outward-paid cover of

$$0 \leq t \leq 0.2, \qquad 1/11 \leq \rho \leq 1$$

has minimum lower bound

$$V_{2.88}(p) \geq 0.005770490891026778. \tag{6}$$

The computation pays $10^{-10}$ per cell for floating roundoff, far below the displayed margin. The gain term is monotone, so the same exclusion holds for every $g \leq 2.88$.

**Theorem 1** (Two-sector exclusion)**.** *At every supplied $g \leq 2.88$, no negative competitor exists in either*

$$\max_i p_i \leq 0.14824732498259885 \quad or \quad \max_i p_i \geq 0.8.$$

*Any unobserved negative competitor must lie in the intermediate band.*

This is not a global transition theorem: the intermediate band remains eleven-dimensional and is not covered.

### 3.3 Transversality of the known branch

ST342 certifies a unique reflection-even stationary tube for

$$g \in [2.902496471747767, 2.9024964917477667].$$

Along a stationary branch, the envelope theorem gives

$$\frac{d}{dg} V_g(p(g)) = -\frac{1}{2} q(g)^T A q(g).$$

Outward evaluation on the complete tube gives

$$q^T A q \in [1.3425510931, 1.3425627340], \qquad \frac{dV}{dg} \in [-0.6712813670, -0.6712755465].$$

Hence the branch crossing is transverse and unique inside the tube. A third global competitor is not excluded.

## 4 ST435–ST436: exact orbit sizes and the Morse shadow

The full dihedral group has order 24. For each of the nine ST421 roots, the center was averaged over its numerical stabilizer and a new invariant radius-$2.1 \times 10^{-9}$ Krawczyk box was certified. Uniqueness inside an invariant box proves that the exact root has the declared stabilizer. Every nonstabilizing transform is separated by at least 0.1373 in one coordinate, many orders of magnitude more than the box diameter.

| representative | O0 | O1 | O2 | O3 | O4 | O5 | O6 | O7 | O8 |
|---|---|---|---|---|---|---|---|---|---|
| stabilizer order | 2 | 24 | 2 | 4 | 2 | 4 | 2 | 2 | 2 |
| orbit size | 12 | 1 | 12 | 6 | 12 | 6 | 12 | 12 | 12 |
| Morse index | 0 | 0 | 1 | 1 | 1 | 2 | 1 | 2 | 2 |

**Proposition 1** (Certified nonexhaustive census)**.** *The nine local representatives generate* 85 *distinct stationary points of the supplied* $g = 4$ *functional. Their index counts are*

$$m_0 = 13, \qquad m_1 = 42, \qquad m_2 = 30.$$

Their partial Morse polynomial and alternating sum are

$$M_{\text{partial}}(t) = 13 + 42t + 30t^2, \qquad M_{\text{partial}}(-1) = 1. \tag{7}$$

The value one equals the Euler characteristic of the simplex. This is a striking internal consistency check, not a completeness proof: further critical points can occur in index-canceling families, and boundary-stratified Morse bookkeeping has not been completed.

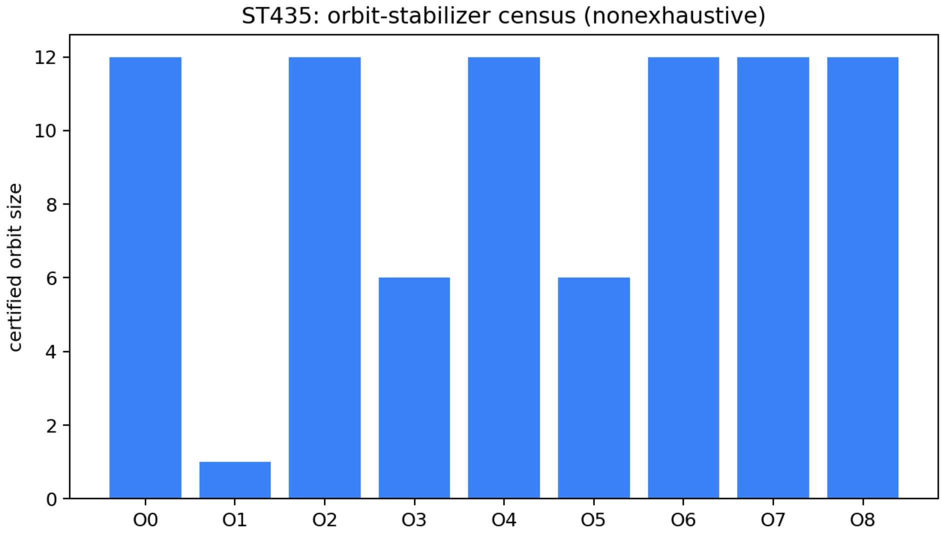


Figure 1: Certified orbit sizes for the nine isolated representatives. The figure does not assert that the atlas is exhaustive.

## 5 ST437–ST438: an enlarged singular attachment and local degree

The compactification remains

$$b = 1/y_3, \qquad c = ay_3, \qquad a = bc,$$

with exponentially flat tails extended by zero at $b = 0$. A common five-box for $(y_1, y_2, c, \nu, T)$ is centered at the $b = 0.0085$ root. Its radii are chosen from the endpoint variation and then verified, rather than trusted, by the Krawczyk inclusion. On the complete interval $b \in [0, 0.017]$ the smallest component margin is

$$1.7121949701 \times 10^{-7} > 0.$$

In the box-scaled infinity norm, the associated derivative contraction is $0.159774 < 1$, which separately pays uniqueness throughout the tube.

**Theorem 2** (Extended compactified attachment)**.** *For every $b \in [0, 0.017]$, the regularized five-equation system has exactly one root in the displayed common box. At $b = 0$ it is the certified limiting face; for $b > 0$ it is the original finite-$y_3$ root. This extends the prior $b \leq 0.002$ result by a factor 8.5.*

The same fixed-box construction fails at $b = 0.018$. That failure is not a fold, band loss, or no-go; it only marks the limit of this box template.

For $b \leq 0.015$, interval left multiplication by the midpoint inverse gives

$$\sup \|I - CJ\|_\infty \leq 0.9381006921 < 1.$$

All Jacobians in the box are therefore nonsingular and homotopic through nonsingular matrices to the midpoint Jacobian, whose determinant is $-0.5667433846\ldots$.

**Corollary 1** (Local degree)**.** *The unique compactified branch component has local Brouwer degree $-1$ for every $b \in [0, 0.015]$.*

This is a local component theorem. It does not exclude additional components outside the box and it does not provide a dimensional infrared scale.

# 6 ST432 and ST439: invariant-algebra progress and the exact boundary

The degree-eight candidate family contains 2028 formal products in an invariant space of dimension 1892, so at least 136 relations are forced. At each of the primes 1000003 and 1000033, ST432 evaluates all candidates at 128 exact mean-zero points, projectively normalizes every nonzero signature, and hashes the complete signatures. There are no zero signatures and no projective duplicates.

**Proposition 2.** *No rational degree-eight syzygy in the declared candidate family has support one or two.*

Indeed, a primitive rational two-term relation reduces nontrivially modulo either prime and would force proportional signatures. The proposition gives a sparsity obstruction; it does not find any of the support-three-or-higher relations.

For the joint odd/even presentation, both primes give decomposable ranks

$$\operatorname{rank} R_5^{\text{dec}} = 72, \qquad \operatorname{rank} R_6^{\text{dec}} = 310.$$

A nonzero modular minor is a nonzero integer minor, hence supplies a lower rank bound over $\mathbb{Q}$. Candidate counts supply upper bounds. Because there are exactly 72 degree-five candidates, the characteristic-zero rank is exactly 72 and the primitive quotient dimension is

$$124 - 72 = 52.$$

At degree six there are 326 candidates, so only

$$39 \le \dim Q_6^{\text{primitive}} \le 55$$

is proved over characteristic zero. Agreement of two primes does not prove the missing upper rank bound.

# 7 ST440–ST443: design, transfer, nonlinear basin, and rank seven

## 7.1 A one-exchange design improvement

The frozen pool has 1800 synthetic mean-zero observations and 365 sextic coefficients. Starting from the prior 500-row pivoted-QR design, the weakest right-singular direction selects one excluded row and removes the selected row with smallest response in that direction. One exchange changes

$$\sigma_{\min} : 1.64404393 \times 10^{-4} \longrightarrow 1.71457251 \times 10^{-4},$$

an E-improvement factor 1.042899. The A criterion $\operatorname{tr}(X^T X)^{-1}$ improves by factor 1.041606, and the condition number falls from 14758.8 to 14132.5. These are deterministic numerical design facts for a synthetic pool, not a measurement apparatus.

## 7.2 Transfer gap

The certified Birkhoff contraction remains the theorem-level bound. Bilinear Ulam grids of side $21, 31, 41, 51, 61$ give a second/first modulus ratio ending at

$$0.5957429137, \qquad 0.5957401632,$$

with last-step difference $2.75 \times 10^{-6}$. The convergence is strong evidence for the synthetic fixture. A collectively compact approximation error is still missing, so these values are not an infinite-dimensional spectral-gap bracket.

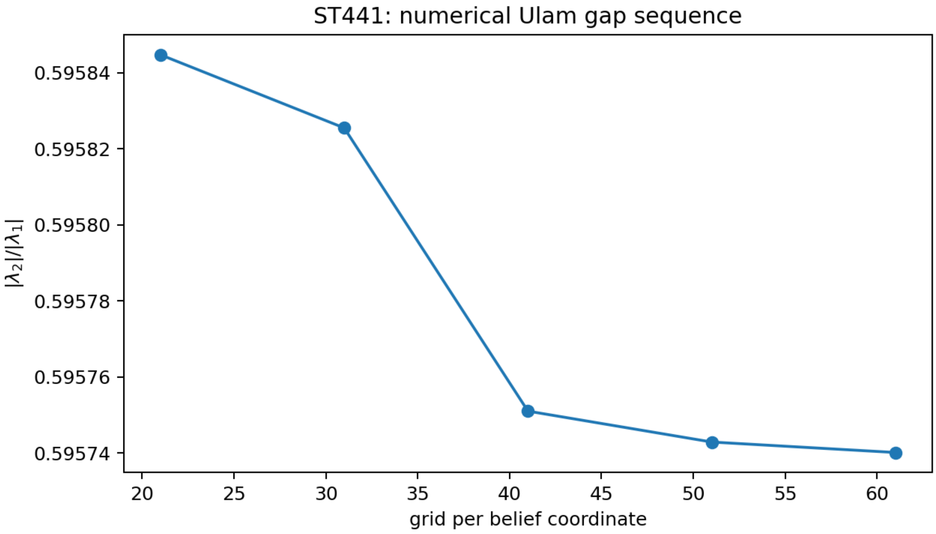


Figure 2: Finite-grid transfer ratios. The plotted convergence is numerical; the cone bound remains the rigorous statement.

### 7.3 Optimized conservative ISS ball

For the localized minimum, use the paid Hessian-variation estimate

$$\mu(R) = \mu_0 - \frac{R}{(p_{\min} - R)^2}.$$

Choosing $R = 1.93 \times 10^{-6}$ yields $\mu \geq 43.69210566$ and admissible tangent forcing

$$\epsilon < \frac{\mu R}{2} = 4.21628819 \times 10^{-5}.$$

Relative to ST427, the certified radius grows by 1.93 and the forcing threshold by 1.309. The result remains local, conditional, and dimensionless.

### 7.4 Rank-seven stop

The rank-seven mediator has the same maximal eigenvalue as $A$ but contains 24 positive and 42 negative off-diagonal pairs. The strict positive-edge water-fill argument is therefore unavailable. No local certified SDP solver is installed, and no independent rational SDP dual was constructed. The global rank-seven variational claim remains blocked; spectral coincidence is not positivity-geometry equivalence.

## 8 What survives falsification

The batch supports four conclusions and no stronger ones.

1. The global transition problem is now localized in probability space, but the intermediate band prevents identification of its critical gain.

2. The singular IR branch is a genuine unique compactified component over a much larger interval and has certified local orientation. Its global topology and physical interpretation remain open.

3. The known stationary landscape is richer than a list of nine numerical points: it contains 85 interval-certified points with an unexpectedly balanced partial Morse census. The balance is not an exhaustion theorem.

4. The invariant algebra has a real characteristic-zero result at degree five and a real sparsity obstruction at degree eight. The full relation basis is still missing.

The deepest surviving interpretation is therefore still operator-theoretic and variational: a supplied strict finite spectral object supports a controlled entropy–quadratic phase family, a compactifiable singular optimization component, and a structured invariant algebra. These are mathematically coherent shadows of a common finite object. They do not yet constitute a theorem from primordial information to dimensional, experimentally testable physics.

# 9 Recommended programs ST447–ST461

| Program | Acceptance test | Priority |
|---|---|---|
| ST447 | Use checkpointed sparse Wiedemann/Lanczos algebra to recover support-$\geq 3$ degree-eight modular relations. Lift across two primes and verify each integer polynomial identity directly. | 1 |
| ST448 | Cover the unresolved band $0.1482473 < \max p_i < 0.8$ at successive gain slabs by interval branch-and-bound with entropy and quadratic relaxations. Accept only a full simplex certificate. | 1 |
| ST449 | If ST448 isolates one global crossing, certify equality, transversality, and exclusion of every third competitor. This is the gate for a unique global critical gain. | 1 |
| ST450 | Construct an interval cover of a full $D_{12}$ fundamental domain for the stationary equations. Determine whether the 85-point census is exhaustive. | 1 |
| ST451 | Build isolating blocks or a Conley connection matrix for the index-one saddles. Do not promote finite-time trajectories. | 2 |
| ST452 | Continue the IR branch beyond $b = 0.017$ with overlapping adaptive boxes and boundary matching; isolate a genuine fold only if interval singularity conditions pass. | 2 |
| ST453 | Complete the IR complement cover, including root ordering, inactive slack, derivative signs, and exponential-tail exclusion. | 2 |
| ST454 | Find exact characteristic-zero degree-six relations, then combine them with ST447 into a minimal joint odd/even presentation through degree eight. | 2 |
| ST455 | Run multi-exchange E/A/D optimization on a frozen training pool and report all metrics on a separately frozen synthetic hold-out pool. | 3 |
| ST456 | Prove a collectively compact Ulam error estimate and convert the transfer sequence into two-sided leading/second spectral-value brackets. | 2 |
| ST457 | Replace the local ISS ball by a certified larger sublevel basin or prove a sharp obstruction caused by the simplex boundary. | 2 |
| ST458 | Implement a sign-aware rank-seven SDP relaxation with a rational dual replay. Accept only a paid bound that closes the localization/curvature gap. | 2 |
| ST459 | Re-run the strict gain-source admission gate only if ST447–ST458 export a genuinely new typed source object; otherwise preserve the block. | Gate |
| ST460 | Re-run selector/QW-2191 only if a nonpremise symmetry-breaking provider is exported; orbit counting alone is inadmissible. | Gate |

| Program | Acceptance test | Priority |
|---|---|---|
| ST461 | Keep the evidence gate closed until an independent record, custody split, frozen hold-out, and preregistered analysis exist. | Gate |

## 10 Reproducibility packet

The executable source is `fin_st432_st446_research.py`. It writes 15 per-program JSON packets, `FIN_ST432_ST446_Results.json`, a CSV ledger, and two figures. The test suite `test_fin_st432_st446.py` contains 15 tests, including live replay of the degree-eight signature audit and the enlarged IR Krawczyk certificate; all pass. The SHA-256 manifest covers the source, packets, ledger, figures, tests, TeX source, and final PDF.

**Epistemic final line.** This release adds proof-grade local mathematics and explicitly bounded numerical evidence. It adds no physical unit, selector, laboratory fact, Standard Model/gravitation derivation, or Theory-of-Everything closure.